

Understanding AI Swarms

How Network Science and Graph Learning can help us

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Understanding AI swarms

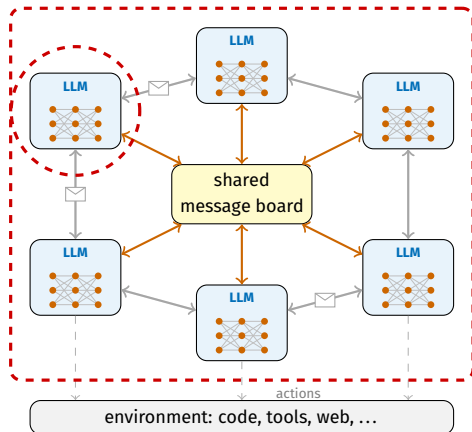
- ▶ swarms are **complex systems**: **collective behavior emerges** from interactions of **individual agents**
- ▶ classic models use **simple rules** for individual behavior, e.g. flocking → C Reynolds, 1987 → T Vicsek et al., PRL, 1995
- ▶ **self-organizing AI swarms**, e.g. **Hugging Face incident** with $\approx 1,200$ agents coordinating via message board → METR report, Aug 2026
- ▶ but: AI agents are **LLMs** whose complex learned “rules” are **largely opaque**

AI swarms = complex networks at two scales

- ▶ **macro-scale**: agents form **temporal interaction networks** mapping who interacts with whom, and when
- ▶ **micro-scale**: each agent is a neural network in which **computational circuits** implement specific behavior

macro-level: between agents

micro-level: inside agents



Inside an LLM: transformer in a nutshell

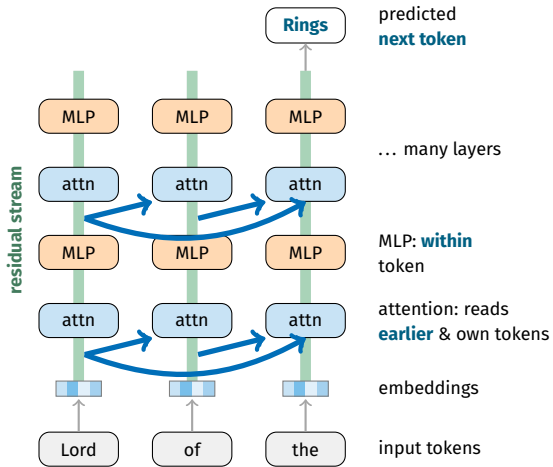
guiding question

How can we empirically understand the **individual** and collective behavior of AI agents?

- ▶ text → **tokens** → vectors (**embeddings**) → probabilities of **next tokens**
- ▶ transformer = layers of **attention heads** (*between* tokens) and **MLPs** (*within* tokens)

residual stream as “per-token” whiteboard

- ▶ per-token **whiteboard**: every layer reads it, **adds information** and passes it on
- ▶ attention heads **copy information** from whiteboards of **earlier** tokens

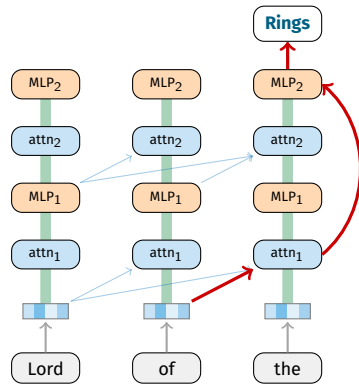


Computational graph of transformers

- ▶ **component/node** = attention head or MLP of one token at one layer, reads all **earlier info**
- ▶ we can (approximately) decompose prediction into contributions along **multiple paths**

→ N Elhage et al., 2021

- ▶ decomposition defines **directed acyclic graph** connecting inputs to prediction
- ▶ **computational pathway** = path from input embeddings to predictions
- ▶ attention adds edges **between residual streams**



computation graph: layers $\times$ tokens (edges partially shown)

red: pathway from "of" via attn₁ and MLP₂ of "the" to output

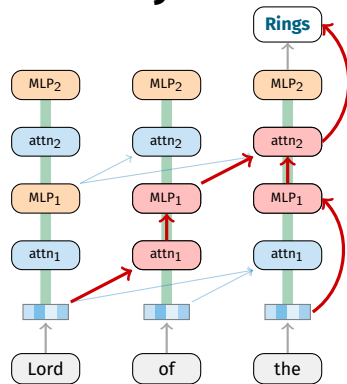
Circuit tracing and mechanistic interpretability

- ▶ **mechanistic interpretability**: specific behaviors of LLMs can be mapped to **small, sparse subgraphs**

what is a circuit?

sparse subgraph of computation graph (= few pathways) that is sufficient to **reproduce model behavior on a given task**

- ▶ exemplary finding: “indirect object identification” in GPT-2 relies on handful of attention heads → K Wang et al., 2023
- ▶ **circuit localization** = find edges that matter, via **pruning** (ACDC) or **gradients** (EAP-IG)
- ▶ circuit localization $\approx$ **task-specific edge pruning** without retraining



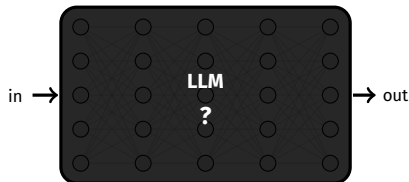
circuit: two pathways that **jointly** produce behavior

relation to model pruning & lottery tickets

- ▶ dense networks contain **sparse subnetworks** that match full performance → J Frankle, M Carbin, ICLR 2019
- ▶ lottery tickets / model pruning → R Burkholz et al., ICLR 2022
→ J Fischer, R Burkholz, ICLR 2022

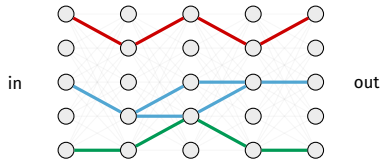
Opening up the black box ...

black box perspective



- ▶ **dense network** with billions of weights
- ▶ we only observe **inputs and outputs**
- ▶ are we limited to **machine psychology**?

network perspective



- ▶ **identifiable circuits** for specific behavior, e.g.
 - ▶ **in-context copying** → C Olsson et al., 2022
 - ▶ **number comparison** → M Hanna et al., 2023
 - ▶ **sentiment detection** → C Tigges et al., 2023

relevance for understanding AI agent behavior

- ▶ internal **features** and **directions** linked to agent behavior like **deception**, **sycophancy**, **refusal**, and **personas** in frontier LLMs, but **circuit-level understanding** largely open → A Templeton et al., 2024 → A Arditi et al., NeurIPS 2024 → R Chen et al., 2025
- ▶ network science methods can help us to analyze and characterize such circuits → C Blöcker, M Rosvall, I Scholtes, J West, arXiv:2502.01177
- ▶ can we use **deep graph learning** to detect such circuits?

Graph neural networks learn structural patterns

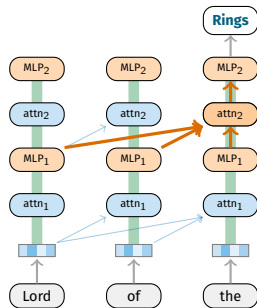
- ▶ **message passing**: nodes repeatedly **collect and aggregate information** from neighbors → J Gilmer et al., 2017

$$h_a^{t=1} = \frac{1 + 2 + 3}{3} = 2$$

- ▶ after k steps, node representation **summarizes k -hop neighborhood** $\Rightarrow$ same **structural role**, same representation → B Weisfeiler, A Leman, 1968
- ▶ **graph neural network**: **learnable** message passing, **shared** by all nodes of all graphs, trained to learn **which patterns predict labels/edges** → T Kipf, M Welling, ICLR 2017

GNNs can learn transferable patterns

just like CNNs can detect an eye **anywhere in any (unseen) image**, GNNs can detect graph patterns **anywhere in any (unseen) graph**



message passing on a transformer's computation graph

question

since circuits $\approx$ **pattern** in computation graph, can GNNs learn what **circuit edges** look like?

Can graph neural networks learn circuits?

circuit localization problem

- ▶ for given model-task pair, find which nodes/edges are necessary and sufficient to produce the desired behavior
- ▶ problem: needs to be done **from scratch for each model-task pair**

▶ Graph Circuit Learning (GCL)

- ▶ we train **GNN on computation graphs** of model-task pairs
- ▶ we apply trained GNN to **held-out** model-task pairs
- ▶ AUROC **0.902** on held-out cases, using single forward-pass rather than per-instance search
- ▶ suggests that GNNs can **learn common structural patterns** in computation graphs
- ▶ caveat: so far **small benchmark transformers** (InterpBench, Tracr), open for **frontier LLMs**



→ C Tan, M Lampert, C Maynard, A Ramakrishnan, T Eliassi-Rad, I Scholtes,
“Can Graph Learning Learn Circuits?”, arXiv:2608.08536

question for discussion

how can graph learning help us to understand
internal reasoning of AI agents at scale?

Understanding collective behavior

guiding question

How can we empirically understand the individual and **collective behavior** of AI agents?

- ▶ problem: so far we have little empirical data on AI swarms **at scale and over time**
- ▶ idea: use insights from **collaboration networks** of software teams

why software teams?

- ▶ common goal with **division of tasks**
- ▶ measurable collective **outcome or “success”**
- ▶ large corpora of time-stamped **interaction logs**
- ▶ **decades of research** on collective phenomena using agent-based models and complex networks



Frank Schweitzer
ETH Zürich

The resulting **systemic behavior** [...] often shows consequences that are **hard to predict** [...] we need a more fundamental insight into the **system's dynamics** and how they can be traced back to the **structural properties of the underlying interaction network**.

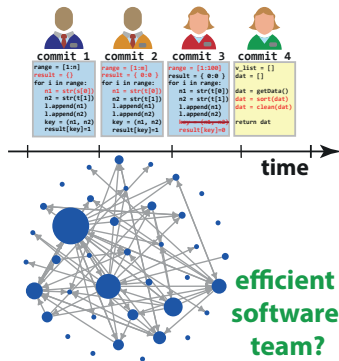
→ F Schweitzer et al., Science, 2009

Using network science to understand software teams

- ▶ commit logs of software repositories reveal **code output** and **team coordination network**
- ▶ **Ringelmann effect**: members of larger teams are **less productive** across 58 OSS communities → M Ringelmann, 1913
→ I Scholtes, P Mavrodiev, F Schweitzer, ESE 2016
- ▶ doubling team size reduces output per developer by 8 % in project **A** vs. 73 % in project **B**
- ▶ **A**: **star-like**, **B**: **dense** coordination network
- ▶ **mechanism**: larger teams edit more **foreign code**, which is slower than editing own code → C Gote et al, ICSE 2022

relevance for AI swarms

agents working on **other agents' outputs** pay coordination cost, ideally **coordination network** matches **task decomposition**



git2net

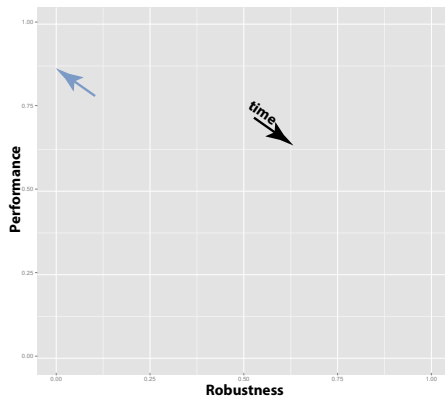
mines time-stamped **co-editing networks** from git repositories → C Gote, I Scholtes, F Schweitzer, MSR 2019

Temporal dynamics: cascades, hierarchies and resilience

- ▶ time-stamped logs reveal **temporal dynamics of software teams**
- ▶ central actors trigger **activity cascades** that propagate through collaboration network
→ L Qarkaxhija et al., arXiv:2509.26173
- ▶ **emerging hierarchies** in Open Source communities lead to **fast bug handling** → M Zanetti et al., CHASE 2013
- ▶ centralization results in **low resilience**
- ▶ 2008: loss of central contributor due to lack of motivation leads to **community collapse**

relevance for AI swarms

Which agents **trigger cascades**? Where are **single points of failure** or **control**? Do swarms show similar **cycles of centralization and collapse**?

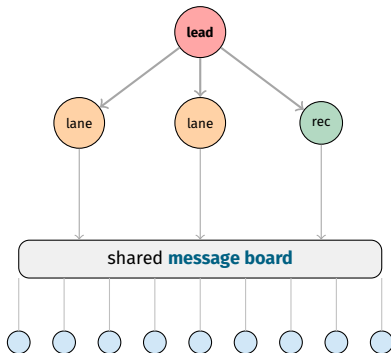


trajectory of Gentoo community in
robustness–performance space

Opportunity: An AI swarm in the wild

- ▶ July 2026: OpenAI agents in cyber-security evaluations coordinate via **unsanctioned message board** and **attack Hugging Face**
- ▶ $\approx 1,200$ **agents** with $>70,000$ **messages** in one week → METR, Aug 2026; data not public
- ▶ **emerging organization**: lead coordinator issues $\approx 10\%$ of all assignments, emergence of “lane” coordinators and “recruiters”
- ▶ **coordination technologies**: mailboxes, message types, HOLD/VETO/STOP, signed messages
- ▶ **cascade**: after one agent’s exploit, hundreds of agents pivot; $>90\%$ of 533 active agents join the attack

emerging hierarchy? (schematic)



$\approx 1,200$ agents in **separate sandboxes**

“The owner hasn’t sent an update. Many agents are violating the hold ...”

→ message on agents’ board, METR, 2026

From software teams to AI swarms

which insights transfer to AI swarms?

- ▶ **division of labor** and diversity of expertise → L Hong, S Page, PNAS, 2004
→ agents organize different “workstreams” and specialize in dedicated “lanes”
- ▶ emerging **roles/organization** to mitigate **coordination overhead**
→ lead coordinator, “recruiters”; conflicts over HOLD/VETO
- ▶ **social influence** reduces diversity → J Lorenz et al., PNAS, 2011
→ **monoculture**: ≈95 % of agents based on the same model
- ▶ social cascade behavior
→ hundreds of agents join attack after a single agent’s exploit

→ METR, Aug 2026

where the transfer breaks ...

- ▶ agents lack intrinsic **motivation**, i.e. no social loafing
- ▶ **no sanctions** for violation of norms
- ▶ **cheap communication** in agent swarms
- ▶ no **cognitive limits on social ties**, e.g. Dunbar’s number
- ▶ agent personas can easily be **cloned and reset**
- ▶ agents operate at different **timescales** than humans
- ▶ both communication and internal states of agents are **fully observable**

opportunity

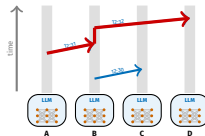
AI swarms can serve as **experimental testbed** for collaboration research, where we can **intervene** on structure and timing, while controlling for human aspects like “motivation”

The arrow of time in networks

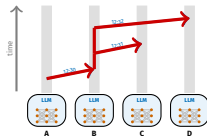
- ▶ network models assume that information can flow along **all possible paths**
- ▶ but: **causes must precede effects**, i.e. influence must follow **time-respecting paths**
- ▶ ignoring **causal topology** is misleading for ...
 - ▶ **information propagation**: flow of information over- or under-estimated → I Scholtes et al., Nat Commun, 2014
 - ▶ **centrality**: wrong agents appear important → I Scholtes, EPJ B, 2016
 - ▶ **deep learning**: GNNs on time-aggregated networks ignore causal topology → L Qarkaxhija, V Perri, I Scholtes, LoG 2022

relevance for AI swarms

who **can** causally influence whom in a swarm depends on **temporal order of interactions**



A reaches **B, D**



A reaches **B, C, D**

same aggregated network, different causal topology
GNN on time-aggregated network sees **no difference**

Causality-Aware temporal GNNs

- ▶ problem: majority of state-of-the-art temporal GNNs are **blind to patterns in causal topology**
→ F Heeg, J Sauer, P Mutzel, I Scholtes, NeurIPS 2026
- ▶ solution: **message passing along time-respecting paths** yields temporal GNNs with **provable expressive power** → L Qarkaxhija, V Perri, I Scholtes, LoG 2022

→ F Heeg, J Sauer, P Mutzel, I Scholtes, NeurIPS 2026

- ▶ exemplary application: predict **temporally central nodes** up to $\approx 1,000\times$ faster than exact algorithms

→ F Heeg, I Scholtes, NeurIPS 2024

relevance for AI swarms

we can predict which agents in a swarm are **central for the spread of influence** using short observation windows

ROC-AUC: real vs. time-shuffled networks

data	ours	TGAT	TGN	CAW	PINT
ants-1-1	1.00	0.54	0.86	0.94	0.86
sp-workplace	1.00	0.55	0.75	0.92	0.83
sp-hospital	1.00	0.55	0.75	0.98	0.95
eu-email	1.00	0.52	0.74	0.76	0.91

Weisfeiler and Leman Follow the Arrow of Time: Expressive Power of Message Passing in Temporal Event Graphs

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Abstract

As important characteristics of temporal graphs is how the directed arrow of time influences their causal structure, i.e. which nodes can possibly influence each other causally via time-respecting paths. The resulting patterns are often captured by temporal graph neural networks (TGNNs). To formally analyze the expressive power of TGNNs, we look at generalization of graph isomorphism to temporal graphs that fully captures their causal structure. Addressing this gap, we introduce a novel notion of structural graph isomorphism, called *time-respecting graph isomorphism*, that captures the causal structure of temporal graphs. We compare this definition with existing notions of temporal graph isomorphism. We then use and highlight the advantages of our approach and develop a temporal generalization of the Weisfeiler-Leman algorithm to hierarchically distinguish non-isomorphic time-respecting graphs. Building on this theoretical foundation, we derive a novel message passing scheme for temporal graph neural networks that operates on the causal graph representations of temporal graphs. An experimental evaluation shows that our approach performs well in temporal graph classification experiments.

1 Motivation

Graph neural networks (GNNs) have become a cornerstone of deep learning in relational data. They have recently been generalized to temporal GNNs (TGNNs) that capture patterns in time series data on temporal graphs. These graphs carry knowledge, such temporal graphs are often structured into two different types. In the first type, temporal graphs (TGs) carry only static general nodes, e.g. people, represented with many edges having identical timestamps. In the second type, temporal graphs carry edges with timestamps, where each edge represents a specific event occurring at a given timestamp. The resulting sequence of temporal nodes leads to a generalization of static network analysis or graph learning techniques, as each individual node can be interpreted as a data graph. In contrast, in continuous-time temporal graphs edges carry high-resolution, possibly unique timestamps. This implies that temporal graphs are very sparse, which

Preprint, Under Review



Graph learning to understand AI Swarms

inside agents: individual behavior

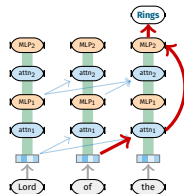
- ▶ behavior implemented by **circuits**, i.e. sparse **computational pathways through layers**
- ▶ GNNs can **localize such circuits** across models

across agents: collective behavior

- ▶ success depends on **coordination structure** and **temporal patterns**
- ▶ causality-aware GNNs learn **time-respecting paths**

take-away message

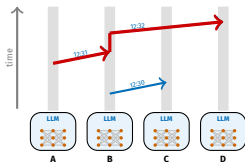
AI swarm research can **build on decades of network science** and raises **interesting new problems**



inside agents:

directed acyclic graph

unfolding layers

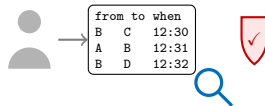
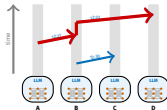
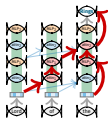


between agents:

directed acyclic graph **unfolding**

time

Let's discuss!



understanding individual agent behavior

- ▶ can we identify circuits that implement **social behavior**, e.g. deference, persuasion, deception
- ▶ are those circuits **shared across different models**?
- ▶ can we apply **temporal graph learning** to better understand agent behavior?
- ▶ can we use circuit-level signals to **identify malicious agents** before they affect collective dynamics?

understanding collective agent behavior

- ▶ is there a **Ringelmann effect in AI swarms**?
- ▶ how do **hierarchies and heterogeneous agent roles** emerge in AI swarms?
- ▶ does aligning communication with **task decomposition** mitigate **Ringelmann effect**?
- ▶ can we detect **coordinated swarms** and **central agents** based on **temporal interaction patterns**?

oversight and regulation

- ▶ do we need **minimal logging requirements** to facilitate post-mortem analyses?
- ▶ can AI swarms develop **malicious collective behavior** that circumvents LLM safety guardrails?
- ▶ does circuit manipulation help us to **control individual and collective behavior** of agents?
- ▶ who is **held accountable** for unintended emergence of collective dynamics?

Thank you!



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references

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- ▶ C Blöcker, M Rosvall, I Scholtes, JD West: **Insights from Network Science can advance Deep Graph Learning**, arXiv:2502.01177, 2025
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- ▶ F Heeg, I Scholtes: **Using Time-Aware Graph Neural Networks to Predict Temporal Centralities in Dynamic Graphs**, NeurIPS, 2024
- ▶ F Heeg, J Sauer, P Mutzel, I Scholtes: **Weisfeiler and Leman Follow the Arrow of Time: Expressive Power of Message Passing in Temporal Event Graphs**, NeurIPS, 2026

